

Detecting Compressed AI-Generated Images via Phase Spectrum Robustness

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Abstract

*This paper aims to present a robust AI-generated image detection framework designed to address performance degradation caused by image compression in online social networks. The key challenges are twofold: 1) compression destroys fragile artifacts that are crucial to existing methods, and 2) it introduces new compression artifacts that interfere with detection. Existing methods typically enhance the compression robustness by collecting original-compression pairs and compression labels. However, the collection and annotation process is highly resource-intensive. To address these issues, we propose a Compression-Robust Phase-Harmonized Transformer, motivated by the observation that **phase spectrum remains stable under compression**. The framework consists of a phase-harmonized cross-modal interaction module that leverages phase spectrum information for feature fusion, enhancing compression robustness, and a multi-domain modulation adapter that further refines fused features while enabling parameter-efficient fine-tuning. In particular, the framework operates **without requiring compression-original data pairs and compression labels**. When limited compression labels are available, we introduce a difficulty-aware consistency loss to maximize their utility by prioritizing hard compressed samples during training, further boosting robustness. Extensive experiments demonstrate that our method significantly outperforms state-of-the-art approaches, exhibiting superior robustness against image compression.*

1. Introduction

Out-of-the-box generative tools [23, 24, 26] have enabled the creation of hyper-realistic synthetic images, increasingly blurring the boundary between authentic and artificial visual content. Although these technologies offer creative potential, they also pose a profound threat to information

integrity and digital trust. Malicious actors can exploit AI-generated content to conduct sophisticated disinformation campaigns, financial fraud, and targeted harassment. As a result, the development of robust and automated detection methods [3, 8, 15, 30, 31, 38] has become not only a technical necessity but also a critical social imperative in combating the growing misuse of synthetic media.

Although satisfactory performance is achieved in the known experiment setting, most existing detectors [10, 19, 20, 39] suffer a sharp performance decline when deployed in real-world environments. This vulnerability primarily stems from their inability to deal with common post-processing operations, particularly the aggressive compression prevalently applied by Online Social Networks (OSNs) [27, 28]. The image compression poses a two-fold challenge for detection models. **First**, it removes the subtle, high-frequency forgery artifacts that are crucial for existing models. As shown in Figure 1 (c), the energy change before and after the compression is mainly reflected in the high-frequency part. **Second**, it introduces new compression artifacts that act as misleading signals. These artifacts can confuse the detector, leading to incorrect classifications and a severe performance drop. As a result, the detector failures occur not only due to the absence of informative forgery cues, but also due to interference from irrelevant patterns, rendering current detectors unreliable in practical settings.

To address these issues, recent approaches [12, 13, 32] have explored training models on compressed data. However, they share a critical limitation: a heavy reliance on large-scale original-compression paired datasets and compression labels. However, in practice, it is often infeasible to obtain both the original, high-quality image and its OSNs-compressed version. Even collecting compression labels is also non-trivial. Consequently, detectors often have access to only a limited number of original-compressed pairs. To this end, this paper aims to develop a compression-robust forgery detection framework that requires neither paired data nor compression labels. Moreover, when limited compression labels are available, we also investigate how to

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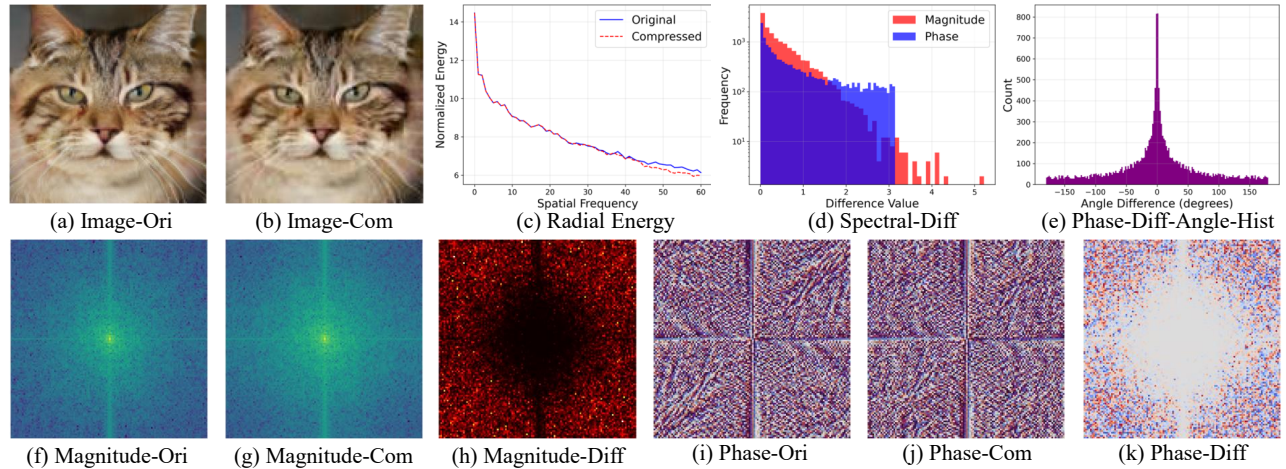


Figure 1. Visualization of the impact of compression on the frequency domain. (a) An original image and (b) its compressed version. (f)–(k) display the amplitude and phase spectra extracted from these images. The “-Diff” denotes the difference between the original and compressed versions. (c) The radial energy distribution plot confirms that energy loss from compression occurs predominantly in the high-frequency regions. (d) The spectral difference distribution reveals that the magnitude of change is significantly larger for the amplitude spectrum than for the phase spectrum. (e) The histogram of phase angle differences shows that for the majority of frequency components, the phase remains unchanged, with differences clustering tightly around zero.

fully leverage limited compression labels to further enhance the compression robustness.

We begin with a foundational analysis of the compression pipeline (detailed in section Method). **Theoretically, an image’s phase spectrum is significantly more resilient to compression artifacts compared to its highly fragile magnitude spectrum.** We empirically validate this hypothesis in Figure 1, where the visualization results confirm that compression-induced distortions predominantly affect the magnitude spectrum, while the phase information remains largely preserved. Building on this critical insight, we propose the Compression-Robust Phase-Harmonized Transformer (CPTFormer), which is designed to exploit this stable phase information as a guiding signal. Specifically, a phase-harmonized cross-modal interaction module (PHCI) enables the robust phase features to guide and fuse the spatial representation learning from the primary RGB stream. To further refine the fused features and adapt the rich visual knowledge embedded in the pre-trained backbone, we introduce a multi-domain modulation adapter (MDMA), enabling parameter-efficient fine-tuning without compromising the model’s rich knowledge. A key advantage of this phase-driven design is its ability to learn generalizable and compression-robust representations using only forgery labels. This contrasts with previous methods, which highly rely on compression labels and paired data.

Despite the label efficiency of our framework, we also introduce a complementary difficulty-aware consistency loss (DACL) to maximally exploit the limited number of available paired samples. Previous methods [13, 32] treat all

compressed samples uniformly, without prioritizing those most susceptible to compression artifacts. Consequently, the detector fails to identify and eliminate the most salient compression distortions. In contrast, the proposed loss function operates dynamically by identifying and prioritizing the most challenging compressed samples, thereby directing the model’s learning capacity toward these critical samples. These compressed samples are most likely to be misclassified. Our approach enables the detector to learn robust features even with limited supervision.

Our main contributions can be summarized as follows:

- We propose the Compression-Robust Phase-Harmonized Transformer (CPTFormer), a novel architecture that, for the first time, systematically exploits the inherent robustness of the phase spectrum for AI-generated image detection. This fundamental principle facilitates the learning of features that are robust to compression perturbations.
- We design the Difficulty-Aware Consistency Loss (DACL), maximizing the utility of limited paired data. Rather than treating all samples uniformly, DACL dynamically prioritizes challenging samples, focusing the model’s learning on the most informative samples.
- We conduct extensive experiments to demonstrate that our proposed framework establishes a new state-of-the-art, significantly outperforming existing methods in realistic, compressed environments and effectively bridging the performance gap between ideal experimental settings and real-world applications.

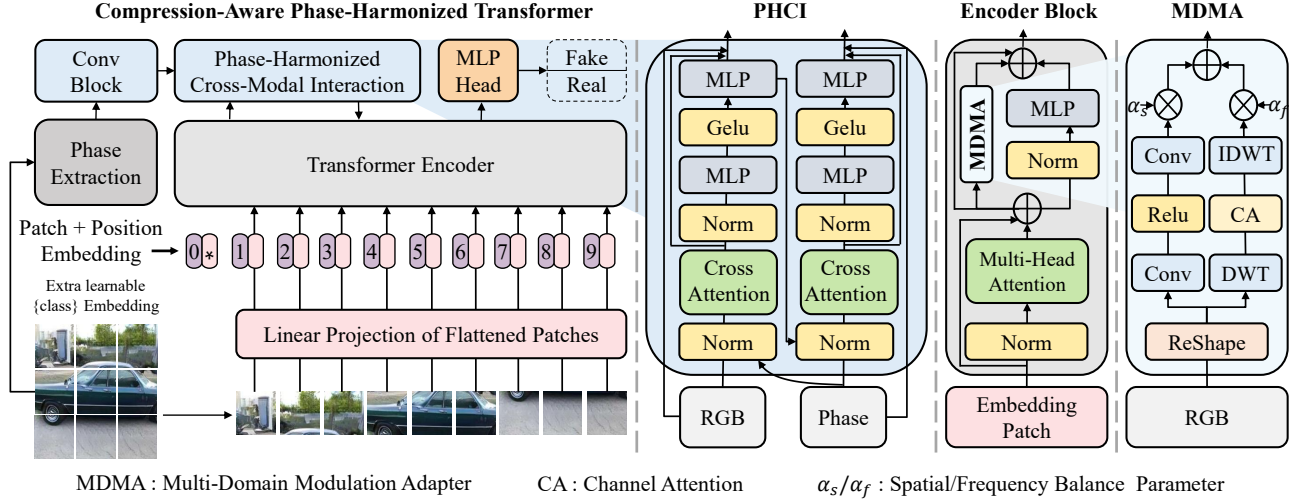


Figure 2. The illustration of the proposed method. Our model employs a dual-branch architecture that integrates standard RGB features from a ViT backbone with robust cues extracted from the phase spectrum. These features are fused via a Phase-Harmonized Cross-Modal Interaction module. For efficient adaptation of the pre-trained model, we introduce a Multi-Domain Modulation Adapter (MDMA), which uniquely refines features in both spatial and wavelet-based frequency domains.

2. Related Works

Initial AI-generated image detection methods focus on spatial [14, 17, 36] or frequency domain [6, 9, 31] statistical discrepancies. A pivotal work [35] demonstrate that CNN-based generators may exhibit shared systematic flaws. Frequency-domain approaches [4, 6] gain attention due to the widespread use of upsampling operations in generative models, which introduce characteristic spectral artifacts. Methods such as FreqNet [31] and F3Net [20] exploit local frequency statistics or high-frequency components as discriminative cues. AIDE [37] introduces a hybrid expert strategy that combines high-level semantic features extracted by CLIP [18, 22] and low-level pixel statistical features [5, 21] based on high and low frequency blocks, attempting to build a more robust defense line from multiple dimensions. Tan et al. [30] introduces the concept of Neighboring Pixel Relationships, highlighting that upsampling operations induce unnatural relationships between adjacent pixels in local regions. Some methods [15, 19, 29] attempt to map images into an unbiased feature space to reduce overfitting to specific artifacts.

Despite significant progress in generalizing to unseen generators, a critical gap remains between performance on clean, academic benchmarks and performance in real-world detection scenarios. This discrepancy is largely attributed to image degradation introduced by Online Social Networks (OSNs). AS the primary distribution channel for synthetic media, OSNs routinely apply JPEG compression to all uploaded images [27, 28]. These operations can severely degrade the subtle artifacts that detectors rely on. Recent

works [12, 13, 32] attempt to improve detector robustness to compression by leveraging paired data. However, it is often infeasible to obtain both the original image and its OSN-processed counterpart. To address this issue, we introduce the compression-robust phase spectrum as an auxiliary map to construct a robust forgery representation.

3. Method

3.1. Problem Formulation

This paper aims to learn a synthesis detector that is robust against compression perturbation, trained on a supervised dataset $\mathcal{D}_{\text{train}}$. This training set includes 80% pristine unpaired images ($\mathcal{D}_{\text{unpaired}}$) and 20% paired subset. The paired subset includes pristine images $\mathcal{D}_{\text{paired}}$ and their compressed versions $\mathcal{D}_{\text{compressed}} = \{C(x) \mid x \in \mathcal{D}_{\text{paired}}\}$, where $C(\cdot)$ denotes a fixed compression operator. The final training set is $\mathcal{D}_{\text{train}} = \mathcal{D}_{\text{unpaired}} \cup \mathcal{D}_{\text{paired}} \cup \mathcal{D}_{\text{compressed}}$.

3.2. Motivation

The widely used JPEG compression [34] algorithm first transforms 8×8 pixel blocks into the frequency domain using the Discrete Cosine Transform (DCT) [2]. This transformation maps spatial pixel values $f(i, j)$ to a set of frequency coefficients $F(u, v)$, where (u, v) denotes the vertical and horizontal spatial frequencies. The information loss primarily comes from subsequent quantization in the compression pipeline. In this step, each frequency coefficient $F(u, v)$ is divided by a corresponding value from the quan-

tization table $Q(u, v)$ and then rounded:

$$F_{\text{quant}}(u, v) = \text{round} \left(\frac{F(u, v)}{Q(u, v)} \right). \quad (1)$$

The quantization table $Q(u, v)$ is designed with small values for low-frequency components and substantially larger values for high-frequency components. **As a result, high-frequency coefficients are aggressively attenuated.** These coefficients often contain subtle artifacts that are critical for many forgery detection methods.

To further examine the implications of this process, we decompose each complex frequency coefficient $F(u, v)$ into its magnitude $|F(u, v)|$ and phase $\angle F(u, v)$:

$$F(u, v) = |F(u, v)| \cdot e^{k\angle F(u, v)}. \quad (2)$$

From this perspective, it becomes clear that the quantization operation directly affects the magnitude component of the coefficient, as below:

$$|F_{\text{quant}}(u, v)| \approx \frac{|F(u, v)|}{Q(u, v)}. \quad (3)$$

In contrast, the phase component $\angle F(u, v)$ remains unaffected by the division, provided the resulting coefficient $F_{\text{quant}}(u, v)$ is non-zero. In such cases, the original phase is preserved: $\angle F_{\text{quant}}(u, v) = \angle F(u, v)$. **Phase information is only lost when the corresponding magnitude is quantized to zero. This scenario primarily occurs in the highest frequency bands, where $Q(u, v)$ values are largest.**

This analysis leads to our foundational insight: the phase spectrum is inherently more robust to JPEG compression than the magnitude spectrum. While magnitude-based features are prone to distortion or elimination, a large portion of phase information survives compression, offering a more stable and reliable signal for detecting forgery artifacts. Our framework is therefore designed to systematically exploit this robust phase information, forming the foundation of our approach to compression-robust forgery detection.

3.3. CPTformer

The proposed Compression-Robust Phase-Harmonized Transformer (CPTFormer) is comprised of a hierarchical phase feature extractor, a phase-harmonized cross-modal interaction module, and is trained with a specialized difficulty-aware consistency loss, as depicted in Figure 2.

Hierarchical Phase Feature Extractor. The extractor is a lightweight convolutional pyramid network that generates a feature pyramid $\{C_l\}_{l=1}^3$ at three distinct spatial scales, corresponding to $1/8, 1/16$ and $1/32$ of the input image resolution. Each feature map $C_l \in h_l \times w_l \times d_l$ captures phase characteristics at a specific level of detail.

To construct a unified representation, each feature map C_l is first projected to the target embedding dimension D via a level-specific 1×1 convolution ϕ_l . The resulting map is then flattened into a sequence of tokens. Crucially, to preserve the vital hierarchical context of each token, we add a learnable level embedding $\mathbf{e}_l \in \mathbb{R}^{1 \times D}$ to tokens from the same scale. The generation of level-aware tokens $\mathbf{p}_l \in \mathbb{R}^{N_l \times D}$ where $N_l = h_l \times w_l$ can be formulated as:

$$\mathbf{p}_l = \text{Flatten}(\phi_l(C_l)) + \mathbf{e}_l. \quad (4)$$

The final phase representation, $\mathbf{p} \in \mathbb{R}^{N_p \times D}$, is formed by concatenating the token sequences from all three scales, providing a comprehensive summary of the phase information for the following cross-modal fusion:

$$\mathbf{p} = \text{Concat}(\mathbf{p}_1, \mathbf{p}_2, \mathbf{p}_3), \quad (5)$$

where $N_p = \sum_{l=1}^3 N_l$ is the total number of phase tokens. This process yields a representation that is not only rich in detail but also explicitly aware of the scale from which each feature originates.

Phase-Harmonized Cross-Modal Interaction. Having extracted token-based representations from both the RGB image ($\mathbf{l} \in \mathbb{R}^{N_l \times D}$) and its phase spectrum ($\mathbf{p} \in \mathbb{R}^{N_p \times D}$), the central challenge lies in effectively fusing these two modalities. To model their complex and complementary relationship, we propose a phase-harmonized cross-modal interaction module (PHCI). This module enables a harmonized fusion process, where the robust phase features guide the refinement of RGB features. Then, the refined RGB features, in turn, enhance the phase representation.

Specifically, the RGB features are enhanced with compression-robust cues from the phase representation. This enhancement is achieved through a cross-attention mechanism, in which the RGB tokens ‘ $\mathbf{l}$ ’ act as queries, while the phase tokens ‘ $\mathbf{p}$ ’ serve as keys and values. The resulting enhanced RGB representation $\mathbf{x}'$ is computed as:

$$\mathbf{l}_{\text{cross}} = \text{CrossAttn}(\text{Norm}(\mathbf{l}), \text{Norm}(\mathbf{p})), \quad (6)$$

$$\mathbf{l}' = \mathbf{l} + \gamma_i \cdot (\mathbf{l}_{\text{cross}} + \text{MLP}(\text{Norm}(\mathbf{l}_{\text{cross}}))), \quad (7)$$

where CrossAttn denotes the standard multi-head attention operation, Norm is Layer Normalization, and γ_l is a learnable gating scalar. This operation allows each RGB token to effectively query the entire set of phase tokens and integrate the most salient compression-robust information before being processed by the main backbone.

After the phase-enhanced RGB tokens $\mathbf{l}'$ are processed through the stack of ViT blocks, they contain contextual semantic features. We denote these fully processed tokens as $\mathbf{l}_{\text{ViT}}$. Subsequently, this rich semantic information in $\mathbf{l}_{\text{ViT}}$ is

utilized to refine the phase representation itself. This is accomplished through another cross-attention mechanism, in which the phase tokens ‘ $\mathbf{p}$ ’ act as queries and attend to the processed RGB tokens $\mathbf{l}_{\text{ViT}}$. The refined phase representation $\mathbf{p}'$ is formulated as follows:

$$\mathbf{p}_{\text{cross}} = \text{CrossAttn}(\text{Norm}(\mathbf{p}), \text{Norm}(\mathbf{l}_{\text{ViT}})), \quad (8)$$

$$\mathbf{p}' = \mathbf{p} + \gamma_e \cdot (\mathbf{p}_{\text{cross}} + \text{MLP}(\text{Norm}(\mathbf{p}_{\text{cross}}))), \quad (9)$$

where γ_p is a learnable gating scalar. This step enables the model to identify the most discriminative phase features in the context of the final detection task, effectively leveraging the global semantic understanding from the RGB stream to enhance the local phase cues. This bidirectional interaction ensures that both modalities are mutually refined, resulting in a more robust and coherent representation.

Multi-Domain Modulation Adapter. The multi-domain modulation adapter (MDMA) consists of two parallel branches: a spatial adaptation branch and a frequency enhancement branch. This spatial adaptation branch follows the conventional adapter design, employing a bottleneck structure to learn specialized feature refinements in the spatial domain. Given features $\mathbf{F}$ processed through multiple stacked interaction modules and ViT blocks, this branch first projects the features down to a smaller bottleneck dimension d_b , and applies a non-linear activation function. Then, we project them back up to the original dimension D . The spatial update $\Delta\mathbf{F}_{\text{spatial}}$ is formulated as:

$$\Delta\mathbf{F}_{\text{spatial}} = \text{MLP}_{\text{up}}(\sigma(\text{MLP}_{\text{down}}(\mathbf{F}))), \quad (10)$$

where $\text{MLP}_{\text{down}} : \mathbb{R}^D \rightarrow \mathbb{R}^{d_b}$ and $\text{MLP}_{\text{up}} : \mathbb{R}^{d_b} \rightarrow \mathbb{R}^D$ are linear projections and σ is a ReLU activation function.

The frequency enhancement branch is designed to explicitly recalibrate artifacts in the frequency domain. First, the input tokens (excluding the class token) are first reshaped into a 2D feature map, $X \in \mathbb{R}^{H \times W \times D}$. We then apply a 2D Discrete Wavelet Transform (DWT), decomposing the feature map into four frequency sub-bands: one low-frequency approximation band (X_{LL}) and three high-frequency detail bands corresponding to horizontal (X_{LH}), vertical (X_{HL}), and diagonal (X_{HH}) details. To amplify these cues within different frequency sub-bands, we apply a channel attention $\mathcal{S}(\cdot)$ to each sub-band independently, which is based on the Squeeze-and-Excitation block [7]:

$$X'_{\text{band}} = X_{\text{band}} \odot \mathcal{S}(X_{\text{band}}), \quad (11)$$

where $\odot$ denotes channel-wise multiplication and the band belongs to $\{LL, LH, HL, HH\}$. The four recalibrated sub-bands $\{X'_{LL}, X'_{LH}, X'_{HL}, X'_{HH}\}$ are used to reconstruct the feature map, X'_{rec} , via an Inverse DWT (IDWT).

Then, the reconstructed feature map is flattened back into a sequence of patch tokens. A zero vector is prepended to align with the class token position, yielding the frequency-domain update $\Delta\mathbf{F}_{\text{freq}}$. The outputs of the two branches are combined via a gated summation to produce the final residual update for the ViT block:

$$\Delta\mathbf{F} = \alpha_s \cdot \Delta\mathbf{F}_{\text{spatial}} + \alpha_f \cdot \Delta\mathbf{F}_{\text{freq}}, \quad (12)$$

where α_s and α_f are two learnable parameters that balance the contributions of the spatial and frequency domains.

Difficulty-aware Consistency Loss. While our method is designed to learn robust representations without requiring paired data and compression labels, we still introduce a difficulty-aware consistency loss (DACL) to efficiently exploit them when such resources exist. The core idea is to up-weight the contribution of the most informative samples, which are the most challenging for the detector. This component, combined with contrastive learning to enforce intra-class consistency, constitutes the difficulty-aware consistency loss.

Let $f_{\theta}(\cdot)$ be the model’s classification head. The cross-entropy loss for the compressed image is $\mathcal{L}_{\text{CE}}(f_{\theta}(x_c), y)$. A high cross-entropy loss signifies a challenging sample. We use this to compute a sample-specific difficulty weight ω :

$$\omega = \exp(\mathcal{L}_{\text{CE}}(f_{\theta}(x_c), y)). \quad (13)$$

Crucially, the gradient is not back-propagated through the calculation of ω . The weight serves purely as a modulating factor on the learning signal, not as a target for optimization itself. The weight grows exponentially with the difficulty, ensuring that challenging samples receive more attention. The difficulty-aware loss can be formulated as below:

$$L_{\text{DAL}} = \omega \mathcal{L}_{\text{CE}}(f_{\theta}(x_c), y) + \mathcal{L}_{\text{CE}}(f_{\theta}(x_o), y), \quad (14)$$

where x_o is uncompressed sample. To further enhance the intra-class consistency and inter-class separability of the learned representations, we employ a contrastive loss. It can be formulated as follow:

$$\mathcal{L}_{\text{con}}^{(i)} = -\log \frac{\sum_{p \in \mathcal{P}(i)} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_p)/\tau)}{\sum_{j \neq i} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_j)/\tau)}, \quad (15)$$

where $\text{sim}(\cdot, \cdot)$ is the cosine similarity, and τ is a temperature hyperparameter. $\mathcal{P}(i)$ is the set of indices of all other samples in the mini-batch that belong to the same class as sample i (positives). The input of classification head is defined as $\mathbf{z}$. The difficulty-aware consistency loss $\mathcal{L}_{\text{DACL}}$ can be written as:

$$\mathcal{L}_{\text{DACL}} = \mathcal{L}_{\text{DAL}} + \lambda_{\text{con}} \mathcal{L}_{\text{con}}, \quad (16)$$

Table 1. The **quality-aware** experimental results on GANs under the **2-class (top)** and **4-class (bottom)** training data setting. **Bold** and underline represent the best and second-best performance. Most comparison results are cited from ODDN [32].

Method	InfoMax- GAN	BE- GAN	Cramer- GAN	Att- GAN	MMD- GAN	Rel- GAN	S3- GAN	SNG- GAN	STG- GAN	Pro- GAN	Style- GAN	Style- GAN2	Big- GAN	Cycle- GAN	Star- GAN	Gau- GAN	Deep- fake	Mean Acc
MesoNet [1]	50.5	50.6	50.0	50.5	50.2	50.4	49.3	50.2	50.9	51.4	51.5	54.2	52.0	53.4	50.6	53.4	50.0	51.2
FF++ [25]	74.4	30.6	75.5	64.2	76.3	61.5	54.9	71.8	82.2	90.4	60.4	65.2	60.5	80.0	74.4	72.3	51.0	67.3
F3Net [20]	65.9	42.6	68.9	55.9	63.7	56.3	53.1	62.1	74.9	84.1	56.7	60.4	56.1	77.8	71.2	68.6	50.4	63.2
MAT [39]	54.5	49.8	59.7	50.1	57.8	50.8	52.8	52.8	56.7	85.7	52.4	53.1	52.9	72.2	57.6	67.6	51.1	57.7
SBI [10]	56.6	51.9	63.4	50.1	59.3	50.6	62.2	52.1	53.0	88.4	51.2	52.4	55.4	74.8	53.6	78.3	51.1	59.3
ADD [12]	52.0	51.0	59.0	50.7	57.2	52.7	44.7	52.3	53.1	70.9	48.0	48.4	51.7	72.4	55.7	64.7	51.3	55.2
QAD [13]	74.8	53.7	79.6	60.1	78.3	66.5	56.0	76.3	80.4	86.3	55.4	57.2	59.1	77.1	79.9	65.8	55.8	69.2
FreqNet [31]	47.7	50.0	50.5	52.6	50.5	47.8	49.1	50.6	51.7	50.1	53.3	55.8	51.6	52.3	51.0	52.1	48.1	50.9
NPR [30]	57.2	53.4	51.8	50.9	50.7	51.8	50.4	49.7	50.8	53.1	52.6	52.3	51.7	55.4	54.3	52.4	49.7	52.2
Fatformer [15]	64.9	53.0	64.5	50.7	61.0	51.7	72.6	55.4	57.9	91.3	54.8	57.7	67.0	87.3	65.1	87.1	51.8	64.3
ODDN [32]	73.1	42.3	76.1	71.2	75.9	72.5	60.5	75.5	85.0	91.3	64.5	69.4	64.3	80.8	78.0	77.3	54.3	<u>71.4</u>
CPTformer	85.8	73.7	87.9	58.6	88.1	61.2	75.3	79.2	80.6	93.2	60.8	60.7	77.7	96.9	86.9	91.3	56.7	77.4
MesoNet[1]	49.5	46.2	52.6	51.3	53.0	53.8	50.4	51.8	54.2	53.3	49.6	53.9	55.1	50.9	52.3	51.7	45.0	51.4
FF++[25]	69.5	26.9	80.3	66.8	79.2	69.9	56.2	75.1	84.4	93.6	62.5	60.8	58.5	80.9	78.5	71.00	52.8	68.7
F3Net[20]	61.0	41.9	65.8	52.9	63.8	55.5	53.8	59.6	71.5	92.2	76.0	59.1	55.9	57.9	71.8	66.0	52.1	62.4
MAT[39]	57.9	46.9	64.2	50.8	63.4	52.4	52.1	56.2	61.8	90.8	54.2	53.9	52.4	73.1	61.4	64.8	51.2	59.5
SBI [10]	60.2	55.7	74.4	50.2	67.1	54.6	61.4	53.0	57.2	96.0	57.4	53.0	55.4	77.6	60.1	74.9	50.6	62.5
ADD [12]	51.7	50.7	57.3	51.3	55.9	52.4	45.2	51.2	52.4	73.5	49.9	50.1	52.2	70.7	54.4	66.4	51.2	55.3
QAD [13]	79.9	37.5	79.5	67.4	76.8	71.7	58.0	79.0	83.5	92.7	64.7	68.7	64.0	81.8	80.3	66.3	52.9	70.9
FreqNet [31]	49.6	50.0	50.9	51.8	49.9	48.6	48.1	50.4	51.6	49.9	56.8	54.7	52.0	49.2	50.3	55.1	49.2	54.2
NPR [30]	57.8	43.2	53.6	65.1	53.9	66.5	53.1	55.7	55.2	73.7	60.2	64.7	60.2	71.2	60.1	59.8	50.8	59.1
Fatformer [15]	70.7	57.9	73.0	51.9	68.5	52.4	73.5	61.7	61.3	98.3	74.2	65.3	67.6	84.2	69.9	87.5	56.9	69.1
ODDN [32]	80.6	38.6	80.7	65.8	78.8	71.1	60.5	76.7	85.8	94.0	67.7	69.9	66.7	84.9	80.5	75.2	54.2	<u>72.6</u>
CPTformer	85.6	74.3	87.4	56.3	86.5	59.5	79.4	77.3	78.3	98.5	76.5	74.9	81.7	94.2	85.8	94.7	56.7	79.3

Table 2. The **quality-aware** experimental results on DMs under the **2-class** training data setting. The first eight models (from ADM to SDv2) are from the DiffusionForensics dataset [36], and the remaining models are from the diffusion test set of Ojha [19].

Method	ADM	DDPM	IDDP	LDM	PNDM	VQDM	SDv1	SDv2	DALLE	Glide- 100-10	Glide- 100-27	Glide- 50-27	ADM	LDM- 100	LDM- 200	LDM- 200-cfg	Mean Acc
MesoNet[1]	54.7	56.5	48.2	50.1	52.6	53.2	43.2	49.3	52.3	56.6	51.4	49.7	52.9	53.5	59.8	51.4	52.2
FF++[25]	62.4	63.8	52.2	50.8	54.4	71.6	58.6	72.8	65.3	61.0	61.7	65.3	59.3	67.6	66.5	64.2	<u>62.3</u>
F3Net[20]	58.5	63.8	50.9	50.4	52.1	50.7	47.8	54.9	64.8	51.8	52.7	52.6	55.8	54.0	53.5	53.9	54.3
MAT[39]	60.7	62.3	49.7	52.6	61.4	60.9	53.3	65.1	67.0	57.5	56.8	58.4	53.1	66.3	66.8	64.0	59.7
SBI [10]	60.2	65.4	51.8	55.7	56.3	62.0	51.6	59.0	59.7	59.9	59.2	60.9	55.7	68.0	68.3	63.8	59.8
ADD [12]	54.9	57.9	47.3	49.6	50.4	57.9	44.3	55.4	54.3	54.6	54.7	64.4	55.3	58.2	57.7	55.9	54.1
QAD [13]	66.8	68.4	57.9	56.5	61.3	58.9	54.3	57.6	63.0	60.4	59.4	62.1	59.7	65.7	66.2	64.6	61.4
FreqNet [31]	55.6	62.1	58.0	49.1	54.6	52.3	47.4	61.4	42.6	46.1	46.5	48.9	56.7	42.8	41.7	42.2	50.5
NPR [30]	55.0	61.8	57.2	67.5	50.8	66.4	52.1	70.9	61.7	62.4	61.9	64.4	50.1	54.1	54.1	57.7	59.3
Fatformer [15]	56.6	70.3	59.9	50.0	72.3	50.8	35.6	50.0	58.9	58.0	57.7	59.4	57.6	70.9	71.7	53.7	58.3
ODDN [32]	58.7	62.5	52.6	51.4	49.5	63.3	45.2	74.1	50.2	57.0	56.5	57.6	59.8	53.3	53.0	52.7	56.1
CPTformer	62.7	74.2	66.7	50.7	78.0	60.4	45.0	51.7	78.9	63.1	63.6	65.6	62.5	83.8	83.5	63.2	65.9

where λ_{con} is a scalar hyperparameter that balances the contributions of the different loss terms. This composite objective guides the model not only to classify correctly but also to learn a well-structured and robust feature space.

4. Experiments

4.1. Settings

Dataset. For model **training**, our model is trained on images generated by ProGAN [11], sourced from the Foren-Synths dataset [35]. To ensure a fair comparison, we follow the data protocol established in ODDN [32]. As mentioned

in problem formulation, the training set is partitioned into two parts: a majority (80%) of pristine, unpaired data and a remaining 20% forming a paired subset, in which each image is paired with a corresponding JPEG-compressed version at a quality factor of 40. For model **test**, we conduct experiments on four diverse test benchmarks: the Foren-Synths [35] dataset and the GANGen-Detection dataset [30] for GANs evaluation, the DiffusionForensics [36] and the diffusion test set of Ojha [19] for DMs evaluation. Two compression evaluation scenarios are employed: **quality-aware**, where the compression ratio is fixed, and **quality-agnostic**, where the compression ratio is randomly selected,

Table 3. The **quality-agnostic** experimental results on GANs under the **2-class (top)** and **4-class (bottom)** training data setting.

Method	InfoMax- GAN	BE- GAN	Cramer- GAN	Att- GAN	MMD- GAN	Rel- GAN	S3- GAN	SNG- GAN	STG- GAN	Pro- GAN	Style- GAN	Style- GAN2	Big- GAN	Cycle- GAN	Star- GAN	Gau- GAN	Deep- fake	Mean Acc
MesoNet[1]	46.3	44.3	59.7	60.0	59.8	58.7	47.8	56.3	69.1	55.0	51.0	49.9	53.9	60.5	64.8	49.9	51.1	53.3
FF++[25]	66.9	37.7	79.4	56.6	77.1	60.5	55.0	69.2	79.8	87.8	55.1	59.8	57.1	79.9	75.6	71.6	52.0	66.0
F3Net[20]	58.0	48.8	61.9	51.5	59.3	53.2	52.0	54.9	61.1	83.4	52.7	54.9	55.0	73.7	65.9	66.7	52.4	59.3
MAT[39]	54.2	49.9	59.6	50.5	57.6	51.2	52.1	52.7	57.8	86.1	52.3	53.0	52.7	70.3	58.2	68.0	51.3	57.7
SBI[10]	56.6	51.9	63.4	50.1	59.3	50.6	62.2	52.1	53.0	88.6	51.3	52.4	55.7	76.0	53.9	78.1	51.2	59.4
ADD [12]	51.8	50.9	59.0	50.7	57.1	52.8	45.0	52.3	52.9	70.2	48.0	48.7	51.8	71.9	55.5	65.1	51.3	57.9
QAD [13]	72.3	55.2	80.0	61.5	78.3	65.5	54.5	76.5	79.2	86.4	56.4	58.0	57.4	82.6	77.8	63.5	56.5	68.3
FreqNet [31]	49.4	59.5	48.8	51.6	49.3	48.5	47.1	50.6	50.6	50.9	55.9	62.3	53.4	51.2	50.0	53.4	48.7	51.8
NPR [30]	58.7	54.8	52.1	53.3	51.8	53.3	50.8	50.2	51.3	53.5	53.8	54.0	53.4	50.8	54.7	52.7	49.7	52.9
Fatformer [15]	65.6	53.4	65.2	50.9	62.1	51.8	74.5	56.0	58.1	92.0	55.3	58.0	67.7	90.0	65.9	87.5	52.0	65.1
ODDN [32]	72.1	44.1	76.8	68.1	76.5	73.3	58.0	75.6	83.5	90.8	61.1	65.9	63.9	83.5	77.0	72.9	55.0	70.7
CPTformer	85.8	71.9	87.6	57.8	87.3	60.8	73.9	79.4	79.9	92.4	59.1	58.5	75.9	95.8	85.8	90.7	55.5	76.3
MesoNet[1]	58.7	45.4	63.5	62.9	62.0	50.2	48.7	58.4	64.1	55.4	52.0	48.1	53.7	63.2	62.0	49.6	51.8	54.3
FF++[25]	68.9	29.9	82.0	63.3	80.4	67.2	55.5	75.4	82.0	93.0	61.1	59.8	57.9	80.1	78.6	67.3	51.9	67.9
F3Net[20]	62.0	43.4	65.8	53.2	64.1	56.7	55.4	58.8	67.7	92.5	76.6	62.3	56.8	60.5	71.0	71.3	51.1	63.4
MAT[39]	52.2	49.3	62.5	50.6	60.3	51.7	53.3	53.9	58.6	92.2	54.4	54.9	54.0	76.5	59.4	68.4	51.0	59.4
SBI[10]	61.3	57.4	74.8	50.3	67.5	54.6	61.5	53.2	57.1	95.9	57.2	52.9	55.4	78.3	59.3	74.6	50.7	62.6
ADD [12]	51.0	50.2	54.4	50.3	53.4	50.7	46.2	50.5	50.9	75.8	51.4	51.6	52.7	72.6	52.3	66.4	50.7	55.0
QAD [13]	76.7	46.4	79.6	68.5	77.1	73.6	58.3	76.3	81.0	90.2	65.3	71.3	64.6	81.8	77.1	66.7	55.1	71.0
FreqNet [31]	49.0	60.4	49.2	51.0	49.8	48.5	46.4	50.2	50.6	50.5	55.6	62.2	53.4	50.0	50.7	53.1	49.3	51.8
NPR [30]	56.6	45.2	52.1	61.9	53.1	67.9	52.9	54.2	53.3	72.8	60.3	65.2	60.3	64.6	60.5	60.8	51.8	58.4
Fatformer [15]	70.6	58.3	71.7	51.6	67.7	52.3	72.8	61.2	61.6	98.2	72.8	64.9	67.3	86.5	69.6	87.2	56.5	68.9
ODDN [32]	80.4	35.1	81.0	68.7	78.2	74.5	62.2	77.5	81.7	91.7	69.2	70.4	68.0	78.8	73.4	73.8	55.3	72.1
CPTformer	84.2	71.7	86.4	54.7	84.6	57.1	78.2	75.9	77.9	98.6	70.9	69.3	79.5	93.5	83.9	94.4	54.5	77.4

Table 4. The **quality-agnostic** experimental results on DMs under the **2-class** training data setting.

Method	ADM	DDPM	IDDPM	LDM	PNDM	VQDM	SDv1	SDv2	DALLE	Glide- 100-10	Glide- 100-27	Glide- 50-27	ADM	LDM- 100	LDM- 200	LDM- 200-cfg	Mean Acc
MesoNet[1]	53.3	60.1	50.6	40.0	48.9	47.7	46.4	52.5	53.8	55.0	48.6	50.9	50.1	49.5	55.8	49.6	50.8
FF++[25]	62.1	66.3	53.8	50.0	55.7	55.1	54.2	66.8	65.2	57.1	56.5	57.0	58.3	60.2	58.9	58.3	58.5
F3Net[20]	54.5	64.3	50.7	49.8	50.4	49.8	40.7	51.3	59.4	51.3	51.8	52.3	52.6	51.8	52.3	51.7	52.2
MAT[39]	59.8	65.5	51.9	54.9	58.1	69.4	50.9	67.4	57.6	55.3	55.1	55.6	57.5	58.1	57.4	60.4	58.4
SBI[10]	55.2	65.9	53.6	53.0	54.4	53.2	42.4	53.2	56.6	55.5	56.9	56.2	54.7	56.8	57.4	57.2	55.1
ADD [12]	52.8	63.6	50.9	52.6	53.5	50.8	41.5	50.6	53.3	50.1	50.0	50.4	51.3	54.3	54.6	56.0	52.3
QAD [13]	60.4	63.7	54.5	56.1	56.2	54.3	52.7	51.9	58.2	58.8	54.6	53.3	52.7	53.7	54.4	62.5	57.4
FreqNet [31]	55.6	62.5	56.7	45.8	52.7	50.4	48.6	60.3	36.8	42.8	42.2	48.2	57.2	34.4	34.3	36.2	47.8
NPR [30]	55.8	65.1	54.2	49.5	68.8	42.5	51.4	54.2	68.6	54.9	55.9	57.0	53.8	61.0	61.3	57.8	56.9
Fatformer [15]	57.9	68.8	58.2	50.0	73.7	50.8	35.8	50.0	60.0	58.9	59.0	60.0	58.7	70.7	70.9	53.5	58.6
ODDN [32]	56.3	63.6	53.3	43.6	51.6	52.7	39.3	61.5	52.6	57.2	56.7	58.8	57.8	52.9	51.6	50.0	53.7
CPTformer	60.9	71.6	62.2	52.0	79.3	56.2	43.9	52.2	78.3	59.8	60.6	60.2	60.6	78.8	78.5	61.1	63.5

representing a more realistic and challenging scenario.

Implementation details. Our framework adopts the pre-trained CLIP-ViT [22] as the backbone architecture. Input images are processed through a standard preprocessing pipeline: first resized to 256×256 pixels, then cropped to 224×224 dimensions. To enhance model robustness, random cropping is employed as a data augmentation strategy during training, while inference is performed using a fixed center crop. The model is trained for a total of 20 epochs. Optimization is conducted using the AdamW [16] optimizer with an initial learning rate of $2e-5$ and a mini-batch size of 32. To evaluate model performance and enable fair compar-

ison with existing methods, classification accuracy (Acc) is used as the primary evaluation metric. The hyperparameter λ_{con} in Equation 16 is set to 0.01.

4.2. Quality-Aware Evaluation

Based on the ProGAN-generated image of ForenSynths dataset [35], we conduct experiments under two training data settings: a 2-class setting (chair, horse) and a 4-class setting (car, cat, chair, horse). Table 1 summarizes the results on datasets generated by GANs. Our method achieves a new state-of-the-art performance with an average accuracy of 77.4% and 79.3%, marking a significant improvement of 6% and 6.7% over the previous best-performing

Table 5. Ablation experiments for the proposed method under the **2-class quality-aware setting**.

Method	InfoMax- GAN	BE- GAN	Cramer- GAN	Att- GAN	MMD- GAN	Rel- GAN	S3- GAN	SNG- GAN	STG- GAN	Pro- GAN	Style- GAN	Style- GAN2	Big- GAN	Cycle- GAN	Star- GAN	Gau- GAN	Deep- fake	Mean Acc
Baseline	60.0	52.5	64.3	54.7	61.6	58.3	66.5	57.5	60.4	86.4	60.2	60.8	64.8	79.2	71.6	80.7	58.8	64.6
+ PHCI	82.8	72.6	85.8	58.5	84.9	62.7	68.1	78.9	81.2	88.7	56.9	56.3	70.3	89.5	87.4	85.4	60.7	74.7
+ MDMA	85.7	71.3	88.1	56.6	87.0	60.4	74.4	78.1	79.7	91.8	59.6	58.3	75.2	93.1	86.1	90.7	56.1	76.0
+ DACL	85.8	73.7	87.9	58.6	88.1	61.2	75.3	79.2	80.6	93.2	60.8	60.7	77.7	96.9	86.9	91.3	56.7	77.4

method. This highlights the strong adaptability of our framework to diverse GAN’s architectures.

The experiment results on diffusion models are presented in Table 2. Our framework achieves superior performance, reaching the highest mean accuracy of 65.9% and exceeding the previous state-of-the-art by 3.6%. This result highlights its strong capability for identifying subtle forgery artifacts across diverse generative strategies. The consistent performance advantage over all competing methods confirms that the robustness of our approach is not confined to GAN-based artifacts but effectively extends to more complex and realistic forgery detection scenarios.

4.3. Quality-Agnostic Evaluation

Under the quality-agnostic setting, all test images are compressed using a randomly selected compression ratio. This setup closely mimics real-world detection scenarios. Tables 3 and 4 present the results on GANs and DMs, respectively. In the 2-class setting, our model achieves a mean accuracy of 76.3% on GANs, outperforming the previous state-of-the-art method by 5.6%. This performance advantage is further reinforced in the 4-class training setting, where our method achieves an average accuracy of 77.4%, demonstrating improved scalability under increased classification complexity. This consistent performance again validates the strong generalization ability enabled by our phase-centric design.

4.4. Ablation Study

To evaluate the contribution of each key component in our proposed method, we perform a comprehensive ablation study. Starting from a baseline model, we gradually incorporate the proposed modules and measure the performance improvement at each stage. The results are summarized in Table 5. The baseline model establishes a reference performance level with a mean accuracy of 64.6%. Then, the introduction of the phase-harmonized cross-modal interaction module (PHCI) allows the model to integrate features from the RGB stream with robust phase information, improving the mean accuracy to 74.7%. Next, we introduce the multi-domain modulation adapter (MDMA) for parameter-efficient fine-tuning. This leads to a further performance gain, with the accuracy increasing to 76.0%. Finally, the inclusion of the difficulty-aware consistency loss (DACL) during training brings the mean accuracy to 77.4%.

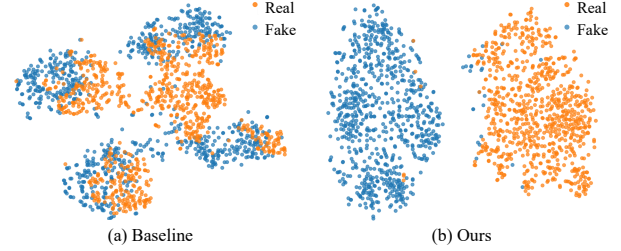


Figure 3. The t-SNE visualization.

The progressive improvements at each step clearly demonstrate that all components work synergistically to achieve the overall state-of-the-art performance of the final model.

4.5. Visualization

To provide an intuitive understanding of why our proposed model achieves superior performance, we visualize the learned feature distributions using the t-SNE technique [33], as shown in Figure 3. It can be observed that the baseline model (CLIP-ViT) produces a feature space where the embeddings for real and fake images are severely entangled. The lack of a clear separation boundary indicates that the model struggles to learn discriminative features, often confusing real and fake samples. In contrast, the feature space learned by our model exhibits a remarkably well-structured distribution. The real and fake embeddings form two distinct, compact, and widely separated clusters. This clear margin demonstrates that our method has successfully learned highly discriminative and robust representations.

5. Conclusion and Limitation

In this work, we address the critical challenge of robust detection of AI-generated images, where the performance of existing detectors is significantly degraded by compression artifacts introduced by Online Social Networks. Our approach is based on a key observation: the phase spectrum of an image exhibits inherent robustness to compression distortions. To exploit this property, we propose the Compression-Robust Phase-Harmonized Transformer, a novel framework specifically designed to integrate and harmonize information from both RGB and phase-domain representations. Extensive experiments show that our method substantially outperforms existing state-of-the-art approaches.

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